

RENNES BRETAGNE AP, France

WMO: 071300

Lat: 48.0689N

Lon: 1.7339W

Elev: 124

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.6	28.3	19.6	14.8	29.7	23.0	17.4	31.2	27.5	50.9	24.9	50.3	4.1	20	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	86.6	68.1	82.7	66.5	79.0	65.2	70.2	82.4	68.2	78.9	66.5	76.0	8.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	96.7	74.6	64.4	91.0	72.0	62.9	86.2	70.4	34.1	82.3	32.4	78.9	31.2	76.3	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
21.4	18.8	16.6	DB	21.2	93.4	3.2	3.9	18.9	96.2	17.0	98.5	15.2	100.7	12.8	103.6	
			WB	20.4	73.1	3.4	2.4	17.9	74.9	15.9	76.3	13.9	77.7	11.5	79.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	42.8	43.7	47.5	51.1	57.1	62.9	66.4	66.0	61.4	55.7	48.2	43.5
	DBStd	10.23	6.74	5.96	5.40	5.33	5.51	5.64	4.97	5.13	4.98	5.45	5.70	6.95
	HDD50	917	232	186	114	48	6	0	0	0	0	14	99	218
	HDD65	4284	687	597	544	416	253	106	42	46	131	290	505	667
	CDD50	2343	10	9	35	82	225	386	508	495	342	191	44	16
	CDD65	234	0	0	0	0	7	41	85	77	22	2	0	0
	CDH74	2220	0	0	0	14	91	443	792	663	207	10	0	0
	CDH80	662	0	0	0	0	11	138	247	222	43	1	0	0
Wind	WSAvg	8.0	8.8	8.9	8.6	8.4	8.1	7.7	7.7	7.2	7.2	7.4	7.8	8.4
Precipitation	PrecAvg	29.6	3.0	2.3	2.0	2.2	2.4	2.0	2.0	1.9	2.3	3.1	3.3	3.1
	PrecMax	38.0	7.0	5.6	6.5	6.4	5.1	5.5	5.3	4.6	6.5	5.5	6.6	7.2
	PrecMin	22.7	0.4	0.1	0.3	0.4	0.9	0.3	0.5	0.3	0.3	0.9	1.2	0.7
	PrecStd	4.8	1.7	1.2	1.3	1.3	1.0	1.3	1.1	1.2	1.6	1.3	1.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	60.6	68.9	76.3	81.2	91.1	91.4	92.8	85.6	74.6	63.1	58.4
		MCWB	54.3	52.6	56.0	60.3	64.9	69.8	69.2	69.6	66.4	63.2	59.0	55.7
	2%	DB	55.7	56.2	63.2	70.1	76.3	83.6	86.2	85.8	79.8	69.8	60.6	56.9
		MCWB	53.1	51.5	53.5	58.4	62.6	66.9	67.5	68.2	64.6	62.1	57.1	54.7
	5%	DB	54.0	54.2	59.3	65.5	72.2	78.8	82.1	80.9	75.4	66.4	58.7	55.4
		MCWB	51.5	50.6	52.3	55.5	61.0	65.1	66.3	66.8	62.9	60.1	55.7	53.4
	10%	DB	52.3	52.6	56.5	61.5	68.0	74.6	78.1	76.7	71.6	64.0	56.8	53.7
		MCWB	49.9	49.6	51.2	53.0	58.5	63.1	65.1	64.6	61.4	59.3	54.0	51.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	54.4	57.4	62.6	67.4	73.1	72.0	72.9	68.9	65.2	60.6	56.8
		MCDB	57.0	57.7	64.7	72.4	78.0	86.8	86.6	87.8	79.6	71.2	62.1	57.6
	2%	WB	53.8	52.8	55.1	58.9	64.2	68.8	69.6	70.2	66.4	63.2	58.1	55.2
		MCDB	55.5	55.3	60.3	68.1	73.3	80.3	82.3	82.1	75.9	67.7	59.9	56.5
	5%	WB	52.1	51.5	53.6	56.3	61.7	66.2	67.6	67.9	64.6	61.6	56.1	53.7
		MCDB	53.8	53.7	58.1	63.6	70.3	76.7	78.5	77.6	72.3	65.2	58.3	55.3
	10%	WB	50.4	50.0	52.1	54.2	59.5	63.9	66.0	66.1	62.9	60.1	54.3	51.9
		MCDB	52.1	52.5	55.9	60.2	66.8	72.7	75.4	74.0	69.3	63.2	56.5	53.5
Mean Daily Temperature Range	5% DB	MDBR	10.5	12.7	15.4	17.4	17.8	18.9	19.5	19.5	18.7	14.6	12.0	10.7
		MCDBR	10.1	13.9	20.1	24.4	24.4	26.1	27.2	26.5	24.9	16.7	12.4	10.0
		MCWBR	8.7	10.3	12.3	13.5	12.5	12.0	11.1	10.8	11.5	9.9	9.5	9.0
	5% WB	MCDBR	9.9	12.3	16.5	21.3	21.7	23.5	23.3	22.3	20.3	14.4	11.3	9.7
MCWBR		9.1	10.1	10.9	12.5	11.9	11.8	10.7	10.2	10.7	9.7	9.4	9.2	
Clear-Sky Solar Irradiance	taub	0.330	0.346	0.376	0.393	0.400	0.399	0.396	0.385	0.367	0.361	0.336	0.323	
	taud	2.410	2.373	2.285	2.244	2.271	2.308	2.321	2.371	2.400	2.417	2.422	2.386	
	Ebn at Noon	233	255	265	272	274	274	273	272	265	246	228	219	
	Edh at Noon	22	28	35	40	41	40	39	35	31	26	21	20	
All-Sky Solar Radiation	RadAvg	323	587	937	1404	1661	1840	1756	1530	1210	697	398	276	
	RadStd	30	67	106	155	135	187	110	127	92	55	38	34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page